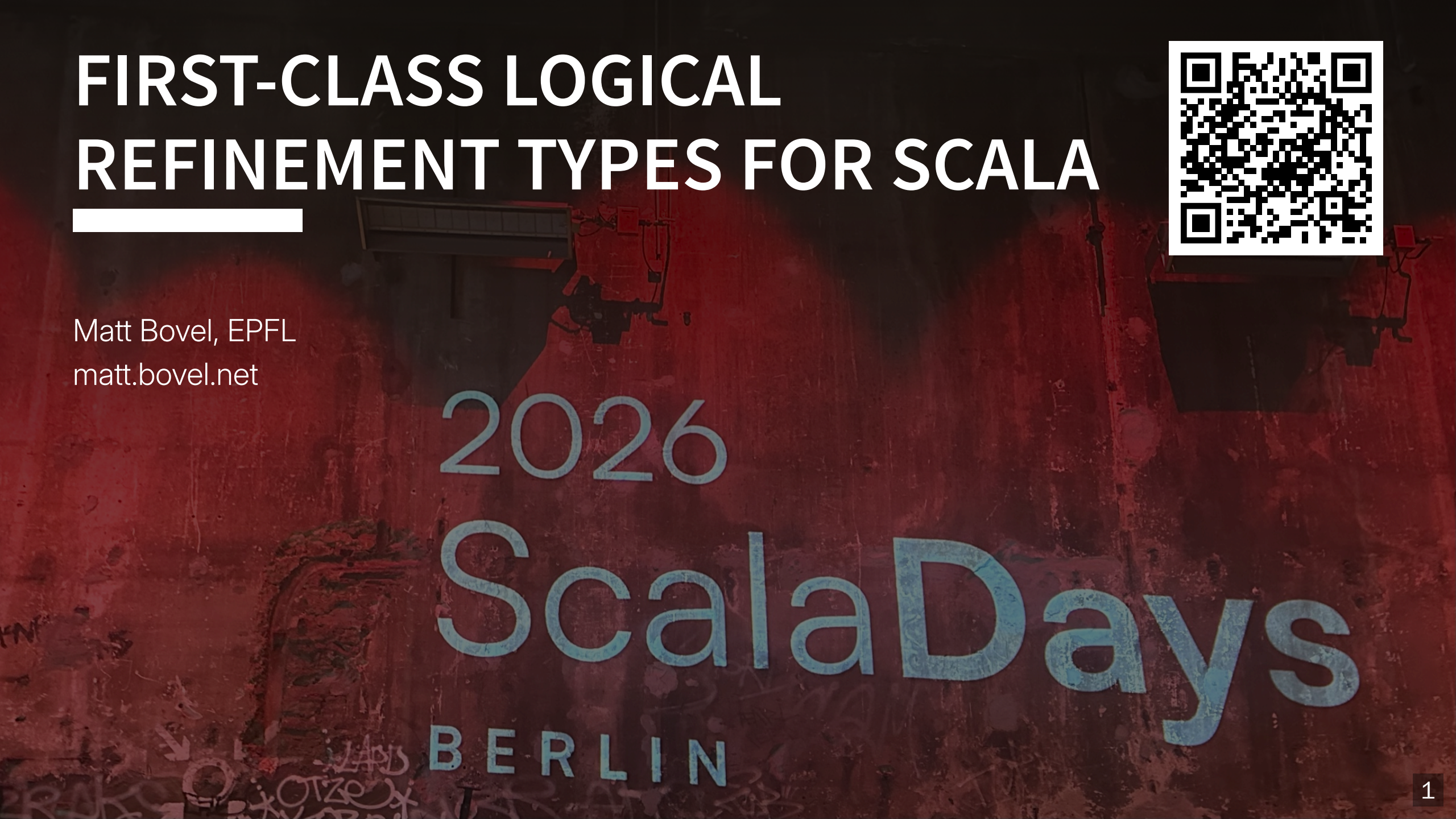
FIRST-CLASS LOGICAL REFINEMENT TYPES FOR SCALA

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WHAT IS THIS ABOUT?

A *prototype* implementation of (logical) refinement types for Scala.

A *refinement type* is a type *refined* with a logical predicate. For example:

```
type Pos = {x: Int with x >= 0}
```

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SYNTAX

Long form, whith an explicit binder:

```
type NonEmpty[A] = {l: List[A] with l.nonEmpty}
```

Short form, reusing a name already in scope:

```
val x: (Int with x % 2 == 0) = 42
// desugars to:
val x: {v: Int with v % 2 == 0} = 42
```

Surveyed and implemented by Quentin Bernet for his Master thesis.

EXAMPLE 1: SIZED LISTS

```
def zip[A, B](
  xs: List[A],
  ys: List[B] with ys.size == xs.size
): {zs: List[(A, B)] with zs.size == xs.size}

def concat[T](
  xs: List[T],
  ys: List[T]
): {zs: List[T] with zs.size == xs.size + ys.size}

val xs: List[Int] = ???
val ys: List[Int] = ???
zip(concat(xs, ys), concat(ys, xs))
```

WHOAMI



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First-Class Refinement Types for Scala

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Refinement types—types qualified with logical predicates—have proven effective for lightweight verification in languages like Liquid Haskell, F*, and Dafny. However, in these systems refinements are either written in a separate specification language or treated as second-class annotations, disconnected from the host language's type system. This disconnect creates usability barriers: programmers must maintain two mental models, and refinements cannot interact with features like type inference, subtyping, or overloading.

We present the design of first-class refinement types for Scala 3, where refinements are ordinary types that participate in subtyping, inference, and pattern matching alongside existing language features. We prove type soundness of a core, pure calculus mechanized in Rocq, combining dependent function types, bounded polymorphism, positive equi-recursive types, union and intersection types, and refinement types, using a fuel-bounded definitional interpreter and semantic typing. A distinctive design choice is our partial-correctness semantics: predicates are arbitrary terms that may diverge, and type soundness requires no termination assumptions. Finally, we implement our design as a prototype extension of the Scala 3 compiler with a lightweight e-graph-based solver for predicate entailment.

CCS Concepts: • **Software and its engineering** → **Compilers**; **Formal software verification**; *Data types and structures*; • **Theory of computation** → **Type theory**; *Program semantics.*

Additional Key Words and Phrases: Refinement Types, Scala, Definitional Interpreter, Semantic Types, Rocq

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1 Introduction

A *refinement type* restricts a base type to values satisfying a logical predicate. For example, the type $\{x : Int \mid x > 0\}$ denotes the positive integers. Logical refinements as subsets of values have a long history [20], going back at least to Cartwright [6]. Freeman and Pfenning [12] coined the term

REFINEMENT VS. ASSERTIONS

Why not use assertions instead?

```
def zip[A, B](
  xs: List[A],
  ys: List[B]
) : List[(A, B)] = {
  require(xs.size == ys.size)
  ???
}.ensuring(_.size == xs.size)
```

1. **Runtime overhead**: assertions are checked at runtime, refinement types are checked at compiletime (by default).
2. **No static guarantees**: assertions are only tested with the inputs you run them with. Refinements types hold *for all values*.
3. **Not part of the API**: assertions are not visible in the function type / scaladoc.
4. **Hard to compose**: assertions cannot be passed as type argument:

```
type Pos = {x: Int with x >= 0}
def foo(l: List[Pos])
def bar(f: Pos => Pos)
```

EXAMPLE 2: FANSI

With assertions:

```
def True(r: Int, g: Int, b: Int) =
  True0(r, g, b, trueIndex(r, g, b))

def trueIndex(r: Int, g: Int, b: Int) =
  require(0 <= r && r < 256)
  require(0 <= g && g < 256)
  require(0 <= b && b < 256)
  r << 16 | g << 8 | b
```

With refinement types:

```
type Comp =
  {v: Int with 0 <= v && v < 256}

def True(r: Comp, g: Comp, b: Comp) =
  True0(r, g, b, trueIndex(r, g, b))

def trueIndex(r: Comp, g: Comp, b: Comp) =
  r << 16 | g << 8 | b
```

FIRST-CLASS?

In our prototype, refinement tyes are ordinary Scala types, participating in inference, **overload resolution** (example 3), **subtyping** (example 4), etc.

This is contrasting with other languages such as Liquid Haskell, where refinement types are implemented as a plugin that runs after type checking:

```
{-@ x :: {v:Int | v mod 2 == 0 } @-}
let x = 42 :: Int in ...
```

Our implementation is directly integrated into the Scala 3 compiler:

```
val x: (Int with x % 2 == 0) = 42
```

ERROR MESSAGES

Predicates are type-checked like other Scala expressions:

```
def f[A](l: List[A] with l.notEmpty) = () // error
```

```
-- [E008] Not Found Error: 6_error_messages.scala:9:27
9 |def f[A](l: List[A] with l.notEmpty) = () // error
  |                         ^^^^^^^^^^
  |      value notEmpty is not a member of List[A] - did you mean l.nonEmpty?
```

ERROR MESSAGES AND INFERENCE

Same inference and error reporting as for other Scala types:

```
type PosInt = {v: Int with v > 0}
def g[T](f: T => Unit, x: T) = f(x)
g((x: PosInt) => x * 2, -2) // error
```

```
-- [E007] Type Mismatch Error: 6_error_messages.scala:17:26
17 |  g((x: PosInt) => x * 2, -2) // error
   |                          ^^
   |                          Found:    (-2 : Int)
   |                          Required: {x: Int with x > 0}
```

EXAMPLE 3: OVERLOAD RESOLUTION

```scala
def min(l: List[Int] with l.isSorted) =
  l.head // O(1)

def min(l: List[Int]) =
  l.min // O(n)

def ex1(l: List[Int] with l.isSorted) =
  min(l) // calls the first overload
```

EXAMPLE 4: BOUNDED POLYMORPHISM

Inspired by Abstract Refinement Types (Vazou et al., ESOP 2013).

```
given Ord[Int] = ...

type Even = {v: Int with v % 2 == 0}

def ex2: Even =
  maximum(List(2, 4, 6))

def maximum[T: Ord, U <: T](xs: List[U]): U =
  xs.reduce(max)
```

REFINEMENT TYPES VS. EXISTINGS CONSTRUCTS

Scala already have *structural refined types*:

```
def zip[A, B](
  as: List[A],
  bs: List[B] {
    val size: as.size.type
  }
): List[(A, B)] {
  val size: as.size.type
}
```

And singleton types, and opaque types, and compiletime operations:

```
opaque type SizedList[
  X <: Int: Singleton,
  A
] <: List[A]

def zip[A, B, Size <: Int: Singleton](
  as: SizedList[Size A],
  bs: SizedList[Size, B]
): SizedList[Size, (A, B)]
```

REFINEMENT TYPES VS. LIBRARIES

With Iron, predicates are defined as type-level constraints:

```
extension (l: List[Int]) def isSorted(): Boolean =
  l.zip(l.drop(1)).forall((x, y) => x <= y)

final class IsSorted
given Constraint[List[Int], IsSorted] with
  override inline def test(inline l: List[Int]): Boolean = l.isSorted()
  override inline def message: String = "List must be sorted"

def min(l: List[Int] :| IsSorted): Int = l.head

min(List(1, 2, 3))                // ⛔ cannot be verified at compile time
min(List(1, 2, 3).refineUnsafe) // ✅ checked at run time
```

LIMITATIONS OF EXISTING APPROACHES

Existing constructs (structural refinements, singleton types, compile-time operations) and libraries (Iron) suffer from:

- *Limited reasoning*: only fields, literals, constant folding, etc.
- *Not inferred*: types must be annotated manually, or programs are not typable at all.
- *Different languages*: predicates are written at the type level (`IsSorted` , `as.size.type`), not at the term level (`isSorted` , `as.size`).

WHAT ARE VALID PREDICATES?

```
var x = 3
val y: Int with y == 3 = x // ⛔ x is mutable

class Box(val value: Int)
val b: Box with b == Box(3) = Box(3) // ⛔ Box has equality by reference
```

The predicate language is restricted to a fragment of Scala consisting of constants, stable identifiers, field selections over `val` fields, pure term applications, type applications, and constructors of case classes without initializers.

Purity of functions is currently not enforced. Should it be?

PURITY OF PREDICATES

By default, functions called in predicates are *assumed* to be pure:

```
class Cell { var field: Int = 0 }
def test(c: Cell) =
  def getAndBump(): Int = { c.field += 1; c.field }
  val v: Int with v == getAndBump() = getAndBump() // ✅ accepted
```

With capture and separation checking, predicates are required to be pure:

```
import language.experimental.{captureChecking, separationChecking}
class Cell extends caps.Stateful { var field: Int = 0 }
def test(c: Cell^) =
  def getAndBump(): Int = { c.field += 1; c.field }
  val v: Int with v == getAndBump() = getAndBump() // ⛔ not pure: uses {c}
```

MIXED-PRECISION TYPE INFERENCE

Refinement types are **not** inferred from terms by default; the wider type is:

```
val x = 42
// x: Int, not {v: Int with v == 42}
```

Why not always infer the precise type?

1. **Backward compatibility:** implicit and overload resolution depend on inferred types; a more precise type changes which instances are found.
2. **Performance:** bigger types, slower comparisons.
3. **Developer eXperience:** users would be flooded with complex types.

Instead, precision is recovered **on demand**, by a standard mechanism called *selfification*, triggered by the bidirectional type-inference algorithm:

```
val x: (Int with x == 42) = 42
```

SOLVER

How does the compiler check `{x: T with p(x)} <: {x: T with q(x)}`?

Need to check that `p(x)` implies `q(x)` for all `x: T`.

Existing implementations in other languages lower to SMT solvers.

Due to packaging and performance concerns, we instead implemented our own **lightweight e-graph-based solver**.

```
{v: Int with v == a && a == b}
  <: {v: Int with v == b}
{v: Int with a == b}
  <: {v: Int with f(a) == f(b)}
```

And domain-specific normalizations such as:

```
{v: Int with v == x + 3 * y}
  <: {v: Int with v == 2 * y + (x + y)}
```

Also beta-reduction, ADT selections reductions and limited reasoning on LIA inequalities.

SUBTYPING: CONSTANT FOLDING

```
{v: Int with v == 1 + 1}      <: {v: Int with v == 2}
```

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SUBTYPING: NORMALIZATION

Arithmetic expressions are normalized using standard algebraic properties, for example commutativity of addition:

```
{v: Int with v == x + 1}     <: {v: Int with v == 1 + x}

{v: Int with v == y + x}     <: {v: Int with v == x + y}
```

Or grouping operands with the same constant factor in sums of products:

```
{v: Int with v == x + 3 * y} <: {v: Int with v == 2 * y + (x + y)}
```

SUBTYPING: LOCAL UNFOLDING

As mentioned, refinement types are not inferred from terms by default. However, the solver can unfold definitions of local `val` (only), even when they have an imprecise type:

```
val x: Int = ...
val y: Int = x + 1

{v: Int with v == y} =:= {v: Int with v == x + 1}
```

SUBTYPING: EQUALITY REASONING

Transitivity of equality:

```
{v: Int with v == a && a == b} <: {v: Int with v == b}
```

Congruence of equality:

```
{v: Int with a == b}           <: {v: Int with f(a) == f(b)}
```

This is implemented using an E-Graph-like data structure.

SUBTYPING: WITH OTHER SCALA TYPES

Literal types are subtypes of singleton refinement types:

```
3 <: {v: Int with v == 3}
```

We plan to support subtyping with other Scala types in the future.

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RUN-TIME CHECKS

Pattern matching branches on whether the predicate holds at run time:

```
val idRegex = "[0-9a-f]{4}-[0-9a-f]{4}"
type ID = {s: String with s.matches(idRegex)}

"a2e7-e89b" match
  case id: ID => ... // holds, id has type ID
  case _      => ... // does not hold
```

RUN-TIME CHECKS: `.runtimeChecked`

You can also use `.runtimeChecked` (SIP-57) when you expect the check to always pass:

```
val id: ID = "a2e7-e89b".runtimeChecked
```

Desugars to:

```
val id: ID =
  if ("a2e7-e89b".matches(idRegex)) "a2e7-e89b".asInstanceOf[ID]
  else throw new IllegalArgumentException()
```

Note: like with other types, you can also use `.asInstanceOf[ID]` directly to skip the check altogether.

`.runtimeChecked` for refinement types was implemented by Valentin Schneeberger.

GRADUAL TYPING

By default, a refinement that cannot be verified statically is a type error:

```
def getInt(): Int = 1
val x: {v: Int with v == 1} = getInt() // ⛔ type error
```

A language import turns these errors into something else:

```
import language.experimental.qualifiedTypes.runtimeChecks // or warn, or silent
```

Mode	Unverified refinement becomes
`runtimeChecks`	run-time check, throws if it fails
`warn`	warning, no run-time check
`silent`	accepted, no run-time check

EXAMPLE 5: MEMOIZED FIBONACCI

A map whose values are related to their keys by a predicate `p`:

```
case class DepMap[K, V](p: (K, V) => Boolean):
  def put(n: K, v: V with p(n, v)): Unit = ???
  def get(n: K): Option[{res: V with p(n, res)}] = ???
```

A function whose result type is its own definition:

```
def fib(n: Int): {r: Int with r == (if n <= 1 then 1 else fib(n - 1) + fib(n - 2))} =
  if n <= 1 then 1 else fib(n - 1) + fib(n - 2)
```

EXAMPLE 5: MEMOIZED FIBONACCI (2)

A memoized version, checked to always return `fib(n)`:

```
val cache: (DepMap[Int, Int] with (cache == DepMap[Int, Int]((k, v) => v == fib(k)))) =
  DepMap[Int, Int]((k, v) => v == fib(k))

def fibMemo(n: Int): {res: Int with res == fib(n)} =
  cache.get(n) match
    case Some(res) => res
    case None =>
      val res: (Int with (res == fib(n))) =
        if n <= 1 then 1 else fibMemo(n - 1) + fibMemo(n - 2)
      cache.put(n, res)
      res
```

EVALUATION

Compilation overhead, against each system's own unchecked baseline:

System	Overhead
First-class (ours)	**0–13%**
Schmid and Kunčak	20–41%
Stainless	≥ 53%

When the feature is unused, the full Dotty CI passes unchanged: the compiler itself (≈200 000 LoC),
≈10 000 tests, a 50-project community build.

≤ 4% slowdown on the official benchmark suite.

FUTURE WORK

Implementation:

- **Better Solver** for what our lightweight solver cannot do.
- **A termination checker**, needed only for the termination-sensitive entailment rules, never systematically for every function in a predicate.
- **Term-parameterized types**, to modularize predicates:

```
type Range(from: Int, to: Int) =
  {v: Int with v >= from && v < to}
```

Theory:

- **Rules for semantic implication.** We leave implication purely semantic; the solver's normalization rules are not yet verified against it.
- **Enforcing purity by default.** Today a function called in a predicate is *assumed* pure, unless capture and separation checking are enabled.
- **Classes and objects**, absent from the presented core calculus.

HOW TO USE

Try it online in the playground.

Scala CLI: add these directives and run `scala --cli-version 1.18.0 run hello.scala`:

```
//> using scalaOrganization ch.epfl.lara
//> using scala 3.10.1-RC1-bin-20260903-e1f9361-NIGHTLY
//> using options -language:experimental.qualifiedTypes
```

sbt: in `build.sbt`:

```
scalaOrganization := "ch.epfl.lara"
scalaVersion := "3.10.1-RC1-bin-20260903-e1f9361-NIGHTLY"
scalacOptions += "-language:experimental.qualifiedTypes"
```

CONCLUSION

We showed:

1. A **prototype implementation** of refinement types in Scala 3 as *first-class* types; normal Scala types that participate in subtyping, inference, overloads resolution and pattern matching.

2. A core **calculus** proven sound in Rocq by semantic typing. It includes refinements, dependent functions and pairs, sums, unions, intersections and equi-recursive positive types, and allows predicates to diverge.

Slides and paper:

matt.bovel.net/scala-refinement-types/



Un type raffiné,
by Marina Granados Castro

BONUS SLIDES

BACKUP: LH USABILITY BARRIERS

From "Usability Barriers for Liquid Types" [1]:

- 4.2 Unclear Divide between Haskell and LiquidHaskell:
  - "comments are usually seen as just optional information in the code and not something that is directly used by the compiler"
  - "It's sort of like you're doing two things at once because you're implementing in Haskell. But you're also talking to GHC, but you're also talking to LiquidHaskell."
- 4.7 Unhelpful Error Messages
  - "[...] error messages produced from typing errors inside the predicates, seemed indistinguishable from those produced by verification errors."
- 4.8 Limited IDE Support
  - "[user] tried to use the function `length`, but since it was not imported, it was impossible to use in this case."

[1] Catarina Gamboa, Abigail Reese, Alcides Fonseca, and Jonathan Aldrich. 2025. Usability Barriers for Liquid Types. Proc. ACM Program. Lang. 9, PLDI, Article 224 (June 2025), 26 pages. doi:10.1145/3729327

BACKUP: `List.collect`

Scala type parameters are *erased* at runtime, so we cannot match on a `List[T]`.

However, we can use `.collect` to filter and convert a list:

```
type Pos = { v: Int with v >= 0 }

val xs = List(-1,2,-2,1)
xs.collect { case x: Pos => x } : List[Pos]
```

BACKUP: VALUE INTERPRETATION (MORE)

Value interpretation of **function types**:

$$\mathcal{V}[\![\Pi x{:}A.\, B]\!]^{\rho}_{\delta}(v) \triangleq \exists \rho_f, b.\, v = \langle \rho_f, \lambda x.\, b \rangle \wedge \\ \forall v_a.\, \mathcal{V}[\![A]\!]^{\rho}_{\delta}(v_a) \implies \mathcal{E}[\![B]\!]^{\rho_f[x \mapsto v_a]}_{\delta}(b)$$

Value interpretation of **recursive types**?

$$\mathcal{V}[\![\mu X.\, A]\!]^{\rho}_{\delta}(v) \triangleq \mathcal{V}[\![A]\!]^{\rho}_{\delta[X \mapsto \mathcal{V}[\![\mu X.\, A]\!]^{\rho}_{\delta}]}(v)$$

$$\mathcal{V}[\![\mu X.\, A]\!]^{\rho}_{\delta}(v) \triangleq \forall n.\, F^n(v)$$

$$F^0(v) = \top, \quad F^{n+1}(v) = \mathcal{V}[\![A]\!]^{\rho}_{\delta[X \mapsto F^n]}(v)$$